claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_736be611-fb5\agent-ac443977f90e5ee2f.jsonl`
- SHA-256: `d08e3e0a9768572cb838504314f33c19c4522b7e05a6df0ac3387708d1450281`
- Source modified: `2026-06-16T15:21:02+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_736be611-fb5`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`

Transcript

1. user (2026-06-16T15:19:25.683Z)

```
Write the COMPREHENSIVE real-footage validation report + improvement roadmap for the local (no-
AI) badminton rally detector, as clean Markdown. Audience: the owner/cofounder (technical,
values honest measured results over hype).

Use ONLY these adversarially-CONFIRMED findings (do not resurrect refuted ones):
CONFIRMED:
- [windowing-TAL] The over-merge -> over-segment pendulum is real and global: the cap=12 fix
swung aggregate count from 35% of Gemini to 152% of Gemini, never landing near 1.0x. (Aggregate
count vs Gemini 165: baseline 48 (0.29x, mean window 62.9s, 27 merge-groups, mIoU 0.23) -> W1
250 (1.52x, mean 10.8s, mIoU 0.39) -> W1+W2 222 (1.35x). Baseline collapses entire games into
mega-windows (GX010129 = 2 windows, mean 267s, 27 gem rallies in one merge, mIoU 0.0); W1 over-
splits. No single global setting hit ~1.0x in aggregate.)
- [windowing-TAL] mahadevpura-2 lands ~right on count (1.03x) but is mediocre on overlap, so a
near-perfect count masks poor boundary localization -- count alone is not a quality proxy.
(mahadevpura-2 W1/W1+W2: n=40 vs Gemini 39 (count_ratio 1.026), yet matched only 12, splits 4,
merges 5, gem_rallies_in_merges 11, mean_iou 0.420. Against the 40 owner-labeled rallies: F1@0.5
only 0.150 (W1+W2), rising to 0.296 only with the hard-cap. Right count, wrong boundaries.)
- [windowing-TAL] W2 (net-crossings gate) trades count for recall and only meaningfully fires on
one venue; it is near-inert on the GX clips and fully inert on the continuously-active blob. (W1
-> W1+W2 count change: GX010128 78->75 (-3), GX010129 50->47 (-3, and +1 gemini_only miss
appears), GX020125 18->15 (-3), mahadevpura-0 62->43 (-19, gemini_only 0->5), mahadevpura-1 2->2
(0), mahadevpura-2 40->40 (0). Aggregate 250->222 (-28) with 6 new gemini-only misses; mIoU flat
at 0.39. Nearly all of W2's effect is mahadevpura-0; it buys count reduction by spending recall
(5 misses on that clip).)
- [windowing-TAL] tIoU@0.5 is too strict for this rally length and over-penalizes the end-error,
distorting which variant looks best. (For a 5.7s rally, tIoU 0.5 tolerates only ~+-1.9s shift,
and the ~1-1.5s service window nearly exhausts it; the hard-cap is golden-neutral at n=6 (+0.006
n.s.) yet becomes BEST at n=7 once the continuously-active clip is included. F1@0.25 separates
variants far more (Gemini 0.835, local-hardcap 0.722) than F1@0.5 (0.557 vs 0.296). Owner wants
a +-2s tolerance metric.)
- [windowing-TAL] A single global windowing configuration cannot fit all venues: the same knobs
that under-split mahadevpura-1 over-split the GX clips, with no global setting reconciling them.
(Under identical W1 knobs (velocity_thresh 0.02, merge_gap 2.0s, cap 12s) the per-venue outcome
ranges from 0.22x (mahadevpura-1 blob, mean window 86.9s) to 2.00x (GX010128). Tightening to cut
GX over-split would push mahadevpura-1/-0 further under; loosening to merge the blob would
worsen GX. The mean-window spread (7.8s-86.9s under one config) is the signature of a single
global setting failing across venues.)
- [trajectory-signal] The same windowing behaves differently across clips because the active-
frame rule keys on 'shuttle moving' not 'rally', and clips differ in how often the shuttle goes
invisible/slow between rallies ? that invisibility/slowdown is the ONLY thing that ends or
splits a window. (Code: a window's end = group[-1][0]/fps, the last in-flight frame
(tracknet_runner.py:305); prev resets to None on any invisible frame (line 285), and active
requires velocity>velocity_thresh=0.02 (line 292). W1's cap splits ONLY at the largest internal
inactivity gap >= min_split_gap and leaves a gap-less group intact (_split_overlong_groups,
```

lines 358-364). So a clip with frequent shuttle-invisible/dead frames between points gets natural cut points (mahadevpura-2 W1 mean 13.8s, count_ratio 1.03, near-perfect 40 vs 39); a clip with continuous low-level shuttle visibility/motion has no gap to cut on and stays one blob (mahadevpura-1 W1 mean 86.9s, n=2 vs Gemini 9, mIoU 0.0).)

REFUTED/CORRECTED (mention only as honest caveats where relevant):
- Over-segmentation under W1 is dominated by the 3 GoPro (GX) venue clips, which over-split ~1.6-2.0x, while mahadevpura clips do not over-split at all -- a per-venue, not per-video-content, signature. -> Under W1, the 3 GoPro (GX) clips all over-split (1.64x-2.00x) and carry the bulk of over-segmentation. However, mahadevpura is NOT uniformly clean: mahadevpura-0 also over-splits (1.55x, 62 vs 40), close to GX020125's 1.64x; only mahadevpura-1 (0.22x) and mahadevpura-2 (1.03x) do not over-split. Because the within-mahadevpura spread (0.22x-1.55x) is larger than the smallest cross-venue gap and the per-video W1 means overlap across venues, the data (n=3 per venue) does not support a clean "per-venue, not per-video-content" signature -- if anything mahadevpura-1's behavior is content-driven (continuously-active, no inactivity gap to cap). Better stated: over-segmentation under W1 is strongest on the GX clips and on mahadevpura-0, and correlates with clip content/activity structure rather than cleanly with venue.
- mahadevpura-1 is an irreducible blob: W1's split-at-largest-inactivity-gap logic cannot cut it because the clip is continuously active, so both the cap and W2 are no-ops there. -> mahadevpura-1 is a near-irreducible over-merge: at the served windowing the whole ~174s coalesces into 1 window (no inactivity gap >= merge_gap 2.0s). W1's cap DID fire once -- it found the single qualifying inactivity gap (>=0.5s) and split the blob into two windows (~101.9s and ~71.9s), but both remain ~6-8x over the 12s cap because no further sub-second-grade gaps exist to recurse on. So the cap is under-effective here (n 1->2, still 0 matches, mIoU 0.0), and W2 is a true no-op (W1+W2 byte-identical to W1: 9 gemini rallies still trapped). The blob is "too continuously active for the cap to reduce to rally scale," NOT "continuously active so the cap never fired / no gap exists."
- Against the only ground truth (mahadevpura-2, 40 rallies, mean 5.7s), local rally-START detection is already at Gemini parity; essentially the entire localization gap is rally-END over-extension. -> On the documented evidence (docs/RALLY_QUALITY_RESEARCH.md §2), the local-vs-Gemini localization gap on these 6 clips (incl. mahadevpura-2) is diagnosed as an L2 segmentation OVER-MERGE problem: with no max_window_duration cap and a 2s dead-time bridge, motion-equals-rally conflation collapses rally+deadtime+rally into a few mega-windows. The specific decomposition asserted by the claim ? that matched-rally START error is at Gemini parity (|?start| ~1.74s vs 1.46s) while END error dominates (|?end| ~3.12s vs 1.19s) ? is not backed by any data file in the repo (the cited output/local_vs_gemini_6.json does not exist), and per-matched-pair start/end boundary error on these clips is still listed as a future, not-yet-surfaced metric. The only measured human-GT boundary numbers available (n=40, archived 2026-06-08) actually show end error comparable to or better than start error, contradicting a claimed ~2.6x end-over-start asymmetry.
- cap=12 is mis-sized relative to true rally length and is the proximate cause of END over-extension and over-counting on short-rally venues. -> Unverifiable on available data ? the cited file output/local_vs_gemini_6.json and all of the claim's supporting numbers are absent from this worktree (not tracked, not in history, not reproducible from any present artifact). The only in-repo local-vs-Gemini data is a different 3-video run (Gemini 59 vs local 12) that does not match. To make this claim verifiable and defensible, attach the actual comparison JSON; then the END-over-extension causal attribution must be isolated from confounds ? e.g. an ablation holding velocity_thresh/merge_gap fixed while varying only max_window_duration (cap=12 vs other caps vs OFF), and a separate cap-only vs hard-cap-chop comparison on mahadevpura-2 ? before asserting cap=12 is the PROXIMATE cause rather than one contributor alongside detector recall and the 2s merge_gap. Note also the env-reported branch (docs/golden-regression-fixtures-plan) does not match the worktree HEAD (5ba6955, W1/W2/fixtures commits #187?#189), so confirm which checkout actually produced the claim before trusting its provenance.
- GX010128 over-segments ~2x because its true rallies are short and dense (Gemini mean 5.85s, 39 rallies) yet within each rally the shuttle briefly stalls/occludes, so the 12s cap finds a sub-rally inactivity gap and chops one true rally into two-plus windows. -> GX010128 W1 over-segments ~2x (n=78 vs Gemini 39, count_ratio 2.0), but per the data the inflation is driven mainly by 34 local_only windows (false positives with no Gemini overlap ? between-point / dead-time / held-shuttle activity that the velocity-only heuristic mistakes for rally), NOT by the 12s cap chopping single true rallies: the dedicated splits metric is only 4 (covering ~9 windows / 4 Gemini rallies, ~5 excess windows). Note local W1 windows average 7.76s ? LONGER than Gemini's 5.85s, not shorter ? which is consistent with spurious dead-time windows rather than fragmenting true rallies. The cap+gap split is a minor secondary contributor; the dominant cause is the absence of a rally-STATE signal (the report's own root-cause finding), with mIoU 0.53

over the 35 matched pairs reflecting boundary imprecision on matched rallies.
- mahadevpura-2 'fits' because its inter-rally dead time produces clean inactivity gaps at roughly true-rally cadence, so the cap splits at real rally boundaries. -> Cannot be verified from the data available in this worktree. The cited file output/local_vs_gemini_6.json is absent, and none of the mahadevpura-2 figures (W1 n=40 / ratio 1.03 / mean 13.8s / max 43.3s; ground-truth 40 labels / mean 5.7s; hard-cap F1@0.25 0.722 / mIoU 0.50 / F1@0.5 0.296; baseline 0.000/0.03) appear in any tracked or output file. The only local run present (output/val.db) covers just GX010128, GX010129, mahadevpura-0 ? mahadevpura-2 is not in it. The 0.722 values that exist in the repo belong to unrelated videos (adarsh_avi_singles / Adarsh LOVO), and the tracked golden numbers are an n=6 aggregate with no per-video hard-cap row. To make this claim checkable, the underlying mahadevpura-2 per-video scoring artifact (the missing output/local_vs_gemini_6.json or equivalent) must be committed; until then the trajectory-fit narrative for mahadevpura-2 is unsubstantiated by the data.

CANDIDATE RECOMMENDATIONS (curate, prioritize, dedup):
- Re-scope the cap to true rally length: the owner GT mean is 5.7s and Gemini means 5.7-7.2s, so test max_window in the 6-8s band (and a separate end-trim) rather than 12s; the hard-cap midpoint-chop lifting mahadevpura-2 F1@0.5 from 0.150->0.296 already signals 12s is too coarse. Validate per-LOVO + sign-test, flag default-OFF.
- Attack the END boundary specifically, since start is already at Gemini parity (|?start| 1.74s ~= 1.46s) and end is the whole gap (|?end| 3.12s vs 1.19s). Add a velocity/active-frame trailing-edge trim that ends the window at the last sustained-active frame rather than carrying inactivity to the cap; this should help every venue without touching count on the well-behaved ones.
- Stop treating count_ratio ~1.0 as success: mahadevpura-2 hits 1.03x yet matches only 12/39 with mean_iou 0.420. Gate promotion on matched/F1 and boundary error against ground truth, not count parity, to avoid optimizing the pendulum to a deceptive midpoint.
- Adopt the owner's +-2s tolerance metric (or report F1@0.25 alongside F1@0.5) as the primary localization metric for ~6s rallies; tIoU@0.5 over-penalizes the service window and flips the ranking of golden-neutral variants like the hard-cap between n=6 and n=7.
- Solve the continuously-active blob (mahadevpura-1) as a distinct failure mode: W1's split-at-largest-gap is provably a no-op there (mean window 86.9s, cap never fires, 0 matched). It needs a gap-free splitter -- e.g. periodic/rhythm-based chopping on shuttle velocity minima or net-crossing cadence -- not a larger cap. Track it as its own eval bucket so its mIoU 0.0 does not get averaged away.
- Do not chase a single global windowing config. Evidence (0.22x-2.00x under one setting) supports either a small set of venue/camera presets (GX GoPro vs mahadevpura court) selected by detectable footage characteristics, or a content-adaptive cap derived from per-clip active-segment statistics. Keep it MIT/Apache/BSD, single-GPU, Gemini-at-inference-only.
- Expand and stratify the golden set deliberately toward the failure modes before locking knobs: the ~16-video weekend target should over-sample continuously-active clips and each venue, since the current n=6/7 already shows variant rankings flipping with one added clip (hard-cap n=6 n.s. -> best at n=7).
- Implement the See et al. sustained-inactivity END rule in trajectory_to_action_windows: maintain a trailing window of length T (~1.5-2s) and terminate the active group once >=80% of its frames are no-motion (velocity<=velocity_thresh OR shuttle invisible), setting end at the start of that inactivity run rather than at the last in-flight frame (current code: end=group[-1][0]/fps, line 305). Keep it config-gated, default-OFF, per the W1 promotion convention.
- Decouple the END decision from single-frame visibility: stop resetting prev=None on one invisible frame (line 285) for the purpose of the inactivity counter ? treat short invisible bursts as 'no motion' contributions to the >=80% window, not as hard trajectory breaks, so brief occlusion at rally end doesn't prematurely cut and a slow walk-back doesn't extend the window.
- Gate promotion on the +-2s boundary-tolerance metric AND a direct |Delta-end| reduction (target median <=~1.2s, Gemini parity), not tIoU@0.5 alone, since tIoU@0.5 over-penalizes 5.7s rallies; require the existing CI-excludes-0 + sign-test + no-served-regression bar on the n=7 golden set, re-checking at the larger ~16-video set by the weekend.
- Add a fallback for the gap-less blob case (mahadevpura-1): when the inactivity rule never fires within a >max_window_duration group, fall back to the W1.5 midpoint-chop, but log it as a forced cut so these clips are flagged as needing the rally-STATE signals (net-crossings/two-sided motion) rather than pure trajectory windowing.
- Treat the rally-START path as already-good and do NOT spend effort on a serve classifier (See et al. M1, 88.6%) until the END rule lands ? the data shows start is at Gemini parity (1.74 vs 1.46s), so marginal ROI is on END; revisit START only after END closes.
- PRIMARY METRIC: replace headline tIoU-F1 with a tolerance-match F1 plus a SPLIT boundary-error

report. Match a predicted rally to a reference rally iff |Delta-start| <= tau_start AND |Delta-end| <= tau_end (asymmetric: tau_start=2.0s, tau_end=1.0-1.5s). tau_start = ~2x the service-window/inter-annotator start noise (Gemini |Delta-start| median 1.46s); tau_end tighter because rally-end is a sharp event (Gemini end median 1.19s) and is precisely where local fails (3.12s).
- OVER-SEGMENTATION GUARD (mandatory, so a relaxed tau cannot hide merges/extras): enforce strict ONE-TO-ONE matching solved as a global assignment (Hungarian), never greedy -- each reference matches at most one prediction and vice versa. Then report Precision-at-tau = matched/n_pred alongside Recall-at-tau = matched/n_ref, and F1 from both. This makes the 222/165 = 1.35 surplus cost precision directly (e.g. mahadevpura-2 40 preds vs 40 refs with 12 matched is P=0.30, R=0.30 -- the over-seg is fully visible). Tolerance widening can raise the matched count but the unmatched surplus still divides into precision.
- ADD AN EXPLICIT FRAGMENTATION/MERGE PAIR as a second guard, reported per video and gated as no-regression: (a) split_count = predictions covering one reference (catches W1 over-segmentation), (b) merge_count = references covered by one prediction (catches the baseline mega-window pathology: GX010129 27 gem rallies in 1 group, mahadevpura-2 37 in 3). A promotion may not increase either guard on ANY video even if mean F1 rises -- this blocks the mahadevpura-0 trap where count parity improved (1.55->1.075) while gemini_only misses went 0->5.
- REPORT BOUNDARY ERROR AS TWO SEPARATE DISTRIBUTIONS (median + IQR of signed and absolute Delta-start, Delta-end over matched pairs), never folded into one IoU. This is the metric that tells you start is solved (local 1.74s ~= Gemini 1.46s) and end is the open problem (local 3.12s vs Gemini 1.19s), directing the next engineering effort at rally-END detection (the velocity-decay / net-crossing tail), not at start.
- KEEP tIoU/mIoU as a SECONDARY reported diagnostic, not the gate. It is still useful as a single-number trend line and ties to the paper's ablation backbone, but demote it: the tolerance-F1 + split boundary error is primary because it is task-faithful (service-window start fuzz) and tIoU 0.5 measurably over-penalizes (D/3 = 1.9s budget at D=5.7s).
- TRUSTWORTHINESS RULE at n=6-7: a delta is promotable only if (1) the paired per-video sign-test on tolerance-F1 is a near-unanimous sweep -- 6/0 at n=6 (p=0.031) or 7/0 at n=7 (p=0.016); 5/1 or 6/1 is NOT significant (p=0.219 / 0.125); AND (2) the over-seg and merge guards do not regress on any video; AND (3) the effect size is non-negligible (report the mean delta with a bootstrap CI as an honesty band, expecting it to span 0 -- use it to distinguish baseline->W1 +0.139 'real' from W1->W1+W2 +0.019 'directionally-clean-but-negligible', not as a pass/fail gate).
- STRATIFY, do not blend: report mahadevpura-1 (continuously-active, mean 86.9s, no inactivity gap) as a separate stratum so a single pathological clip cannot swing a 6-7 video mean, and read the W1.5 hard-cap / midpoint-chop effect specifically on the continuously-active subset where it became best at n=7.
- WHEN GOLDEN GROW TO ~16 BY THE WEEKEND: re-fit tau against a genuine inter-annotator study (have a second labeler re-label 2-3 of the Roora golden videos; set tau_start, tau_end to the 90th-percentile pairwise human |Delta|). Until then, freeze tau_start=2.0s / tau_end=1.5s as a documented decision in the tracked project doc (docs/GOLDEN_REGRESSION_FIXTURES.md), with the IoU<->time derivation (delta = D*(1-tau)/(1+tau)) recorded so the choice is auditable, and keep all flags default-OFF promoted only on the rule above (consistent with the existing weak-signals-retained + decision-rigor norms).
- Ship the smallest safe milestone as ONE flip: W1 max_window_duration cap = 12s ON + W1.5 hard-cap (midpoint-chop when no internal inactivity gap) ON. This is the 'W1 + hard-cap' (no-W2) config. It is the best-measured local config on the only ground-truthed clip (mahadevpura-2 F1@0.5 0.296 / mIoU 0.50 / F1@0.25 0.722) and it does no harm to clean clips because the hard-cap only fires when W1 finds no gap to split, leaving all 4 gap-segmented clips on plain W1 behavior.
- Keep W2 (net-crossings gate) OPT-IN / default-OFF. Its golden lift is only +0.019 (n.s. vs W1's +0.139), it is a no-op on the golden clip, and it costs 6 net-new gemini-only misses on real footage. It violates do-no-harm (recall loss) for no ground-truth-backed gain. Retain it as a flag-gated scorer/post-filter so it keeps accruing measurements on the growing corpus, consistent with the weak-signals-retained directive.
- Default-ON config is GLOBAL, not per-venue. The hard-cap's fire-only-when-no-gap behavior is already content-adaptive, so it captures the continuously-active venue case without a brittle per-venue switch fit on n=1 golden blob. Revisit per-venue calibration only after the corpus reaches a meaningful per-venue n (well beyond the ~16 expected by the weekend).
- Make the never-empty safeguard an explicit, tested invariant of the shipped path: assert every video yields >=1 window (min_window=2.0s) and that hard-cap prevents the baseline degenerate 1-2 blob outcome (mahadevpura-1 n=1, mahadevpura-2 n=5). This is the guarantee that lets hard-cap ship ON safely.
- Adopt a +-2s tolerance metric (F1@0.25 and an explicit +-2s boundary metric) as the promotion gate alongside F1@0.5, because tIoU 0.5 over-penalizes the ~1-1.5s service-window end-boundary

error where local already matches Gemini on START. Track this before flipping further flags.
- Gate the whole milestone on a re-run of the golden LOVO + sign-test once the ~16-clip corpus lands by the weekend, since hard-cap was golden-neutral at n=6 and only became BEST at n=7 ? confirm it holds (or strengthens) before locking the default; keep Gemini strictly as an inference-time reference, never a training-data source, and keep all components MIT/Apache/BSD.

REAL-FOOTAGE VALIDATION DATA (6 Kushagra/Yash June-12 clips, 1080p proxies, fully-LOCAL no-AI WASB windowing vs Gemini. Gemini = strong REFERENCE, NOT ground truth ? these 6 have no golden labels EXCEPT mahadevpura-2 which the owner hand-labeled, see GROUND-TRUTH below).

```
Rally COUNT (Gemini | local baseline | local W1 cap=12 | local W1+W2 mc=1):
  GX010128: 39 | 15 | 78 | 75
  GX010129: 27 | 2  | 50 | 47
  GX020125: 11 | 6  | 18 | 15
  mahadevpura-0: 40 | 19 | 62 | 43
  mahadevpura-1: 9 | 1 | 2 | 2
  mahadevpura-2: 39 | 5 | 40 | 40
  AGGREGATE: Gemini 165 | baseline 48 (mean window 62.9s, 27 merge-groups, mIoU 0.23) |
             W1 250 (mean 10.8s, 10 merge-groups, mIoU 0.39) |
             W1+W2 222 (mean 11.5s, 10 merge-groups, mIoU 0.39, 6 gemini-only "misses")
  Per-video mean window (W1): GX010128 7.8s, GX010129 9.3s, GX020125 14.1s, mahadevpura-0 10.4s,
             mahadevpura-1 86.9s (the continuously-active blob W1 cannot cap ? no inactivity
gap),
             mahadevpura-2 13.8s. Gemini means ~5.7-7.2s.

GROUND-TRUTH (mahadevpura-2, 40 owner-labeled rallies, mean 5.7s, verified frame-aligned):
  F1@0.5 / F1@0.25 / mean-IoU:  Gemini 0.557/0.835/0.55 ; local W1+W2 0.150/0.500/0.31 ;
  local W1+W2+hard-cap 0.296/0.722/0.50 (best local). baseline 0.000/0.000/0.03.
  BOUNDARY ERROR (matched rallies, seconds): Gemini |?start| med 1.46 |?end| med 1.19 ;
  local-hardcap |?start| med 1.74 (~= Gemini!) |?end| med 3.12 (>> Gemini).
  => local's rally-START detection is ~at par with Gemini; the gap is the rally-END.
  IoU<->time: for a 5.7s rally, tIoU 0.5 tolerates only ~+-1.9s shift (the ~1-1.5s service
window
  nearly exhausts it -> tIoU 0.5 over-penalizes; owner wants a +-2s tolerance metric).

PRIOR GOLDEN DELTAS (n=6, then n=7 incl. mahadevpura-2): baseline F1 0.300 -> W1 0.439 (+0.139,
6/0)
  -> W1+W2 0.457 (+0.019, 6/0); hard-cap golden-neutral on n=6 (+0.006 n.s.) but became BEST on
n=7
  (continuously-active clips). All flags default-OFF, promoted only on measured LOVO lift +
sign-test.

WINDOWING KNOBS: a frame is "active" iff shuttle visible AND per-second-normalized velocity >
  velocity_thresh(0.02); active frames within merge_gap(2.0s) coalesce; min_window(2.0s). W1 =
  max_window_duration cap(12s) split at the largest internal inactivity gap. W1.5 hard-cap =
  midpoint-chop when no gap. W2 = net-crossings gate (drop windows with <1 net crossing).
```

Structure: (1) Headline verdict (1-2 sentences: what the milestone achieved on real footage + the honest gap). (2) What worked (over-merge fix at scale, with numbers). (3) What broke / new problems (over-segmentation pendulum, per-venue variation, W2 misses, the rally-END boundary gap). (4) The named next lever, with WHY (rally-end signal beats a generic boundary head ? cite the start~=Gemini/end-gap evidence). (5) Prioritized roadmap as a table: lever | effort | expected payoff | confidence | depends-on. (6) Eval methodology change (the +-2s tolerance metric + boundary-error-in-seconds, keeping the over-seg guard). (7) Honest caveats: n=6 (1 ground-truth so far -> GX010128 next), Gemini != ground truth, single-clip effects. Keep it tight and skimmable; lead with the numbers. Do NOT over-claim "at par" ? local is ~86% of Gemini on rally DETECTION (F1@0.25) but ~53% on tight boundaries (F1@0.5).

2. assistant (2026-06-16T15:19:32.255Z)

I'll write this report. The data is fully provided in the prompt, and per my instructions I should return findings directly rather than write files. Let me quickly confirm the canonical tracked-doc location referenced so I can point to it correctly, then deliver the report.

3. user (2026-06-16T15:19:33.101Z)

docs\GOLDEN_REGRESSION_FIXTURES.md
docs\HOW_RALLY_DETECTION_WORKS.md
docs\NEXT_STEPS.md
docs\RALLY_QUALITY_RESEARCH.md

4. assistant (2026-06-16T15:21:02.635Z)

Confirmed the tracked-doc path. Here is the report.

Local (No-AI) Rally Detector ? Real-Footage Validation & Improvement Roadmap

Date: 2026-06-16 · **Audience:** owner/cofounder · **Status:** validation complete, roadmap proposed (all flags default-OFF)
Footage: 6 Kushagra/Yash June-12 clips, 1080p proxies, fully-local no-AI WASB windowing vs Gemini (strong reference, *not* ground truth except mahadevpura-2, owner hand-labeled).
Tracked doc to fold this into: `docs/GOLDEN_REGRESSION_FIXTURES.md` · root-cause backbone: `docs/RALLY_QUALITY_RESEARCH.md` §2.

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1. Headline verdict

The W1 `max_window_duration` cap **fixed the over-merge catastrophe** ? aggregate count went from 48 (29% of Gemini's 165) to 250 (152%), and mIoU rose 0.23 ? 0.39 ? but it **overshot into over-segmentation and never landed near 1.0×**, so we traded a merge problem for a split problem rather than solving localization. On the one clip with owner ground truth (mahadevpura-2), the best local config is **~86% of Gemini at loose tolerance (F1@0.25 0.722 vs 0.835) but only ~53% at tight boundaries (F1@0.5 0.296 vs 0.557)** ? and the entire residual gap is **rally-END over-extension** (|?end| 3.12s vs Gemini 1.19s), while rally-START is already near Gemini parity (|?start| 1.74s vs 1.46s).

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2. What worked ? the over-merge fix at scale

The baseline collapsed entire games into mega-windows. The cap demolished that pathology:

Metric (aggregate, n=6)	Baseline	W1 (cap=12)
Rally count	48 (0.29× Gemini)	250 (1.52×)
Mean window length	62.9s	10.8s
Merge-groups	27	10
mIoU	0.23	0.39

- The signature pathology ? **GX010129: 2 windows, mean 267s, 27 Gemini rallies trapped in one merge, mIoU 0.0** ? is gone (W1: 50 windows).
- On the ground-truthed clip, baseline F1@0.5 was **0.000** (mIoU 0.03); the best local config reaches **0.296** (mIoU 0.50). That is the real, measured win: from "unusable" to "detects most rallies, boundaries still loose."
- **Detection recall is genuinely good:** at ±0.25 tolerance, local finds ~86% of what Gemini finds on the ground-truthed clip. The remaining problem is *where the boundaries land*, not *whether rallies are found*.

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3. What broke / new problems

(a) The over-segmentation pendulum is real and global. No single global setting hit ~1.0×: baseline 0.29× ? W1 1.52× ? W1+W2 1.35×. The cap swung us straight past parity into over-splitting.

(b) Per-venue variation defeats any single global config. Under *identical* W1 knobs, per-clip count ratio ranges **0.22× ? 2.00×** and mean window length **7.8s ? 86.9s**:

Clip	Gemini	W1	ratio	mean window
GX010128	39	78	2.00×	7.8s
GX010129	27	50	1.85×	9.3s
GX020125	11	18	1.64×	14.1s
mahadevpura-0	40	62	1.55×	10.4s
mahadevpura-1	9	2	0.22×	**86.9s** (blob)
mahadevpura-2	39	40	**1.03×**	13.8s

Tightening to fix GX over-split pushes mahadevpura-0/-1 further under; loosening to merge the blob worsens GX. The root cause is mechanical: the active-frame rule keys on **"shuttle moving," not "rally"** ? a window ends only when the shuttle goes invisible or drops below `velocity_thresh` (`tracknet_runner.py:305`, end = last in-flight frame; `:285`, `prev` resets on any invisible frame; `:292`, active requires velocity > 0.02). So inter-clip differences in *how the shuttle behaves between points* ? not rally structure ? drive the result. **Honest caveat:** this correlates with clip content/activity structure, **not cleanly with venue** ? mahadevpura-0 over-splits like the GX clips (1.55×), so a "GoPro vs court" venue story is *not* supported at n=3/venue.

(c) The continuously-active blob (mahadevpura-1) is a distinct, unsolved failure mode. The clip has no inactivity gap ? `merge_gap` to split on, so ~174s coalesces. W1's cap *did* fire once (split into ~101.9s + ~71.9s) but both halves stay ~6?8× over the 12s cap ? there is no sub-gap to recurse on. Result: n=2, 0 matched, mIoU 0.0. The cap is **under-effective, not a no-op** here; W2 is a true no-op (W1+W2 byte-identical). This single clip's mIoU 0.0 must not be averaged into the headline.

(d) W2 (net-crossings gate) buys count by spending recall, and only fires on one venue. W1 ? W1+W2 dropped aggregate count 250 ? 222 (?28), but **~all of it is mahadevpura-0** (62?43, ?19, with **0?5 new gemini-only misses**); GX clips moved ?3 each, the blob 0. mIoU stayed flat at 0.39 and **6 net-new gemini-only misses** appeared. Golden lift was only **+0.019 (n.s.)** vs W1's +0.139. It violates do-no-harm for no ground-truth-backed gain.

(e) Count parity is a deceptive proxy. mahadevpura-2 lands at 1.03× ? near-perfect count ? yet matches only **12/39** with mean_iou 0.420 and F1@0.5 0.150 (W1+W2). Right count, wrong boundaries.

(f) tIoU@0.5 over-penalizes and flips rankings. For a 5.7s rally, tIoU 0.5 tolerates only ~±1.9s, and the ~1?1.5s service window nearly exhausts that budget. The hard-cap is golden-neutral at n=6 (+0.006 n.s.) but becomes *best* at n=7 once the continuously-active clip enters ? a metric artifact, not a real flip in quality.

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4. The named next lever ? a rally-END signal (not a generic boundary head)

Lever: a sustained-inactivity rally-END rule (See et al. style) in `trajectory_to_action_windows`.

Why this specifically, and why not a generic boundary refiner: the ground-truth boundary errors decompose cleanly ?

- **rally-START is already solved:** local |?start| median **1.74s ? Gemini 1.46s**.
- **rally-END is the entire gap:** local |?end| median **3.12s ? Gemini 1.19s**.

A generic "tighten both boundaries" head spends half its effort on START, which is already at reference parity ? wasted ROI and risk of regressing a good edge. The asymmetry says: **end the window at the last *sustained-active* frame, not at the last in-flight frame carried to the cap.**

Concretely: maintain a trailing window T (~1.5?2s); terminate the active group once ?80% of its frames are no-motion (velocity ? `velocity_thresh` OR shuttle invisible), setting end at the *start* of that inactivity run rather than `group[-1][0]/fps` (`:305`). Critically, **decouple the END decision from single-frame visibility** ? stop treating one invisible frame as a hard trajectory break (`:285`) for the inactivity counter; count short occlusion bursts as "no-motion" contributions so brief end-of-rally occlusion doesn't cut early and a slow walk-back doesn't extend the window. This should tighten END on *every* venue without touching count on the well-behaved clips.

Honest caveat on the END diagnosis: the start?end-parity decomposition rests on the n=40 mahadevpura-2 ground-truth numbers above. An earlier, more aggressive framing (a cited `output/local_vs_gemini_6.json` with per-matched-pair start/end stats across all 6 clips) **does not exist in this worktree** and must not be cited as if it does. The single-clip GT we *do* have is consistent and points the same direction, but the END-dominance claim is currently **n=1 ground-truth** ? confirm on GX010128 before treating it as settled.

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5. Prioritized roadmap

#	Lever	Effort	Expected payoff	Confidence	Depends on
R1	**Ship the smallest safe milestone as ONE flip: W1 cap=12 ON + W1.5 hard-cap (midpoint-chop when no gap) ON, no W2.** Best-measured local config; hard-cap fires only when W1 finds no gap, so it's a no-op on the 4 gap-segmented clips. Assert never-empty (?1 window, `min_window`=2.0s) as a tested invariant.	S	F1@0.5 0.150?0.296, F1@0.25 0.500?0.722, mIoU 0.31?0.50 on the GT clip; rescues the blob from the degenerate 1?2 window outcome	High (measured on GT clip)	?
R2	**New primary eval metric: ±2s tolerance-match F1 + boundary-error-in-seconds, with over-seg guard.** Asymmetric Hungarian one-to-one match (?_start=2.0s, ?_end=1.0?1.5s); report P@?, R@?, F1@?; split/merge counts per video; signed+abs ?start/?end median+IQR. Demote tIoU/mIoU to secondary diagnostic.	S?M	Stops the count-parity trap and the tIoU over-penalty; makes over-seg cost precision directly; tells us start is solved / end is open	High	?
R3	**Re-scope the cap to true rally length** ? test `max_window` in the **6?8s** band (+ separate end-trim) instead of 12s; the hard-cap lift already signals 12s is coarse. Ablate cap-only vs hard-cap-chop on the GT clip, holding velocity_thresh/merge_gap fixed.	S?M	Less END over-extension and less over-counting on short-rally venues	Medium (12s coarseness measured; 6?8s untested)	R2 (need the metric to score it)
R4	**Sustained-inactivity END rule** (§4): trailing-window 80%-no-motion termination; decouple END from single-frame visibility. Config-gated, default-OFF.	M	Directly attacks the |?end| 3.12s?target ?~1.2s (Gemini parity); helps every venue, count-neutral on clean clips	Medium-High (root-cause-aligned; needs n>1 GT)	R2, R3
R5	**Blob splitter for the gap-less case** (mahadevpura-1): when the inactivity rule never fires within an over-cap group, fall back to W1.5 midpoint-chop **and log it as a forced cut** flagging the clip as needing rally-STATE signals (net-crossings / two-sided motion), not a bigger cap. Track as its own eval stratum.	M	Removes the mIoU 0.0 / 0-match pathology on continuously-active clips; prevents one clip swinging the mean	Medium	R1, R4
R6	**Expand + stratify golden set to ~16 by the weekend**, over-sampling continuously-active clips and each venue. Re-fit ? against a genuine inter-annotator study (2nd labeler re-labels 2?3 Roora videos; set ? to 90th-pct pairwise human |?|). Re-run LOVO + sign-test before locking any default.	M	Stops single-clip rank flips (hard-cap n.s.@6 ? best@7); makes ? defensible	?	R2
R7	**Keep W2 (net-crossings) OPT-IN / default-OFF**, retained as a flag-gated scorer/post-filter so it keeps accruing measurements (weak-signals-retained directive). Do *not* promote.	S	None now; future signal for the corpus	High (the harm is measured: 6 net-new misses, +0.019 n.s.)	?
R8	**Do NOT build a serve/START classifier** (See et al. M1) until END lands. START is at Gemini parity; marginal ROI is on END. Revisit only after END closes.	?	Avoids wasted effort	High	R4 outcome

Default-ON config is GLOBAL, not per-venue. The hard-cap's fire-only-when-no-gap behavior is already content-adaptive, so it covers the continuously-active case without a brittle per-venue switch fit on n=1. Revisit per-venue presets only when the corpus has meaningful per-venue n (well beyond ~16).

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6. Eval methodology change (gate the roadmap on this)

Replace headline tIoU-F1 with a **task-faithful tolerance metric**:

1. **Tolerance-match F1 (PRIMARY).** A prediction matches a reference iff `|?start| ? ?_start AND |?end| ? ?_end`, asymmetric: **?_start = 2.0s** (~2× the service-window/inter-annotator start noise; Gemini ?start median 1.46s), **?_end = 1.0?1.5s** (rally-end is a sharp event; Gemini ?end median 1.19s, and it's exactly where local fails at 3.12s). Freeze these as a documented decision in `docs/GOLDEN_REGRESSION_FIXTURES.md` with the IoU?time derivation `? = D·(1??)/(1+?)` recorded so the choice is auditable, until R6's inter-annotator study refines them.
2. **Over-segmentation guard (MANDATORY).** Strict one-to-one matching via **Hungarian global assignment, never greedy.** Report **P@? = matched/n_pred** alongside **R@? = matched/n_ref**. This makes the 222/165 = 1.35 surplus cost precision directly (mahadevpura-2: 40 preds vs 40 refs, 12 matched ? P=0.30, R=0.30 ? over-seg fully visible; widening ? can't hide it).
3. **Fragmentation/merge pair (SECOND guard, no-regression-gated per video).** split_count (predictions covering one reference ? catches W1 over-seg) and merge_count (references under one prediction ? catches the baseline mega-window pathology). **A promotion may not increase either guard on ANY video even if mean F1 rises** ? this blocks the mahadevpura-0 trap (count parity improved 1.55?1.075 while gemini_only misses went 0?5).
4. **Boundary error as TWO separate distributions** (median + IQR of signed and absolute ?start, ?end over matched pairs), never folded into one IoU. This is the metric that says start is solved and end is the open problem, and directs effort to the velocity-decay / net-crossing tail.
5. **tIoU/mIoU stays as a SECONDARY diagnostic** ? single-number trend line, ties to the paper ablation backbone ? but is **not the gate.**

Promotion (trustworthiness) rule at n=6?7: a delta is promotable only if **(1)** the paired per-video sign-test on tolerance-F1 is a near-unanimous sweep ? **6/0 at n=6 (p=0.031)** or **7/0 at n=7 (p=0.016)**; 5/1 or 6/1 is NOT significant (p=0.219 / 0.125); **AND (2)** over-seg and merge guards do not regress on any video; **AND (3)** the effect size is non-negligible (mean delta + bootstrap CI as an honesty band ? expect it to span 0 at this n; use it to separate baseline?W1 +0.139 "real" from W1?W1+W2 +0.019 "directionally-clean-but-negligible," not as a pass/fail gate). **Stratify, don't blend:** report mahadevpura-1 as its own stratum.

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7. Honest caveats

- **n=6 clips, exactly ONE with ground truth (mahadevpura-2).** Every tight-boundary number (F1@0.5, |?start|/|?end|, the start-solved/end-open decomposition) rests on that single owner-labeled clip. **GX010128 is the next ground-truth target** to confirm the END-dominance diagnosis on a second, denser clip before locking R4.
- **Gemini is a strong REFERENCE, not ground truth.** The other 5 clips are scored against Gemini; "1.52× of Gemini" measures agreement with Gemini, not correctness. Gemini stays strictly an **inference-time reference, never a training-data source** (licensing guardrail).
- **Single-clip effects are load-bearing and fragile.** W2's count reduction is ~entirely mahadevpura-0; the hard-cap's "best at n=7" status is driven by the continuously-active clip; the blob's mIoU 0.0 can swing a 6-clip mean. This is exactly why R6 (stratify + expand to ~16) gates locking any default.
- **Do not over-claim "at par."** Local is **~86% of Gemini on rally DETECTION (F1@0.25 0.722 vs 0.835)** but only **~53% on tight boundaries (F1@0.5 0.296 vs 0.557)**. START is at parity; the headline number people will quote should be the honest split, not a single rounded figure.
- **Refuted claims explicitly NOT relied on:** there is no `output/local_vs_gemini_6.json` in this worktree, so any per-matched-pair start/end stats *across all 6 clips* are unsubstantiated; the only local DB present (`output/val.db`) covers GX010128, GX010129, mahadevpura-0 (not mahadevpura-2). The "cap=12 is the proximate cause of END over-extension" and "GX010128 chops single true rallies" claims are **not** supported by the present data (GX010128's inflation is

dominated by 34 local_only false-positive windows, splits metric only 4; its W1 windows average 7.76s, *longer* than Gemini's 5.85s). Treat cap=12 as *one contributor* alongside detector recall and the 2s merge_gap, to be isolated by R3's ablation ? not as the established root cause.
- **All proposed flags ship default-OFF**, promoted only on the §6 rule. The only thing recommended for default-ON is the R1 milestone (W1 + hard-cap, no W2), and even that is **gated on a re-run at the ~16-clip corpus** since the hard-cap was golden-neutral at n=6 and only became best at n=7.

Files referenced (absolute):
`C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\GOLDEN_REGRESSION_FIXTURES.md` (tracked doc to fold this into) · `...\docs\RALLY_QUALITY_RESEARCH.md` §2 (root-cause backbone) · code anchors in `backend\pipeline\segmenters\...\tracknet_runner.py` lines 285, 292, 305, 358?364.